

CondorBrain / Mamba-2 Institutional Trading System

A Mathematical Monograph

Core Dynamics · Risk Geometry · Loss-Landscape Theory

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Abstract

Source Traceability Notice.

All definitions, equations, and propositions in this monograph are explicitly cross-referenced to the originating source documents listed below. Page numbers and section labels correspond to the author-provided PDFs.

Primary Sources

- *Integrated Implementation Strategy: Synthesizing CondorBrain Architectures* (Jan 14, 2026)
- *Enhancing CondorBrain (Mamba-2) for Institutional-Grade SPY Options Trading*
- *Enhancing the Mamba-2 Neural Network for Advanced Trading and ROI Optimization*
- *Loss Function Specification and Training Error Interpretation* (Jan 11, 2026)

Each chapter header below includes parenthetical citations of the form **[Doc-#, p.X, §Y]** indicating document number, page, and section where the equation or concept first appears.

This monograph presents a rigorous mathematical treatment of the CondorBrain / Mamba-2 institutional trading system. The work is structured as a formal applied-mathematics text and introduces explicit **definitions, propositions, lemmas, and asymptotic results** governing signal dynamics, risk propagation, and optimization geometry.

The objective is not merely descriptive, but *theoretical*: to establish provable scaling laws, stability properties, and geometric interpretations underlying neural-state-space trading systems.

Part I — Core Dynamics

(Primary sources: Doc-1, pp.1–3; Doc-2, pp.1–2; Doc-3, pp.1–4)

Chapter 1. Time, Returns, and Probability Space

[Doc-4, pp.2–3; Doc-1, p.1 §Validation]

Definition 1.1 (Discrete Market Time)

Let

$$t \in \mathbb{Z}_{\geq 0}$$

denote discrete market time indexed by bars of fixed duration Δt .

Definition 1.2 (Log-Return)

Source: Doc-4, p.2, *Mathematical Definition of the Huber Loss* (target definition section) Let $P_t > 0$ be the asset price. The (unscaled) log-return is

$$r_t = \log \left(\frac{P_t}{P_{t-1}} \right).$$

Definition 1.3 (Scaled Training Return)

Source: Doc-4, pp.6–7, *Target Scaling and Error Regime Interpretation* For numerical conditioning, define

$$r_t^{(s)} := \alpha r_t, \alpha = 100.$$

Chapter 2. Embedding and State Space Representation

[Doc-2, p.1; Doc-3, p.1; Doc-1, p.3 §Architecture]

Definition 2.1 (Input Feature Vector)

At time t , the feature vector is

$$x_t \in \mathbb{R}^F.$$

Definition 2.2 (Neural Embedding)

Define the embedding map

$$E_t := \text{LayerNorm}(W_{\text{in}}x_t + b_{\text{in}}),$$

with $W_{\text{in}} \in \mathbb{R}^{D \times F}$.

Definition 2.3 (Continuous-Time State-Space Model)

Source: Doc-2, p.1, *State-Space Model (SSM)* The latent state evolves according to

$$\frac{dh(t)}{dt} = Ah(t) + BE(t),$$

with output

$$y(t) = Ch(t).$$

Definition 2.4 (Discrete Selective-Scan SSM)

Source: Doc-2, p.1; Doc-3, p.1, *Selective-Scan Mechanism* After discretization,

$$h_t = \bar{A}_t h_{t-1} + \bar{B}_t E_t,$$

where $\bar{A}_t, \bar{B}_t$ are **input-adaptive operators**.

Lemma 2.1 (Bounded State Growth)

Assume:

1. $\|\bar{A}_t\|_2 \leq 1 - \eta$ for some $\eta > 0$
2. $\|E_t\|_2 \leq M < \infty$

Then the state norm satisfies

$$\|h_t\|_2 \leq \frac{\|\bar{B}_t\|_2 M}{\eta}.$$

Interpretation: selective-scan gating enforces *contractive memory*.

Part II — Risk Geometry

(Primary sources: Doc-4, pp.6–13; Doc-1, p.2 §Pivot Logic)

Chapter 3. Forecast Error and Price Geometry

[Doc-4, pp.6–9]

Definition 3.1 (Forecast Error)

Let $\hat{r}_t$ predict the return. Define

$$e_t := r_t - \hat{r}_t.$$

Definition 3.2 (Price Error Propagation)

For small returns,

$$\Delta S_t \approx S_t e_t.$$

Lemma 3.1 (Sub-Gaussian Forecast Noise)

Assume forecast errors satisfy

$$\mathbb{E}[e_t] = 0, \mathbb{E}[e_t^2] = \sigma_e^2.$$

Then price errors satisfy

$$\mathbb{E}[(\Delta S_t)^2] = S_t^2 \sigma_e^2.$$

Chapter 4. Delta-Weighted PnL Geometry

Definition 4.1 (Delta-Weighted PnL Noise)

For net delta exposure Δ_{net} , define

$$X_t := \Delta_{\text{net}} \Delta S_t.$$

Lemma 4.1 (Variance of PnL Noise)

$$\text{Var}(X_t) = (\Delta_{\text{net}} S_t)^2 \sigma_e^2.$$

Chapter 5. Drawdown Scaling Law

[Doc-4, pp.12–13]

Definition 5.1 (Cumulative Noise PnL)

$$M_N := \sum_{t=1}^N X_t.$$

Proposition 5.1 (Forecast-Noise Drawdown Scaling)

Source: Doc-4, p.13, *Expected drawdown bound from forecast noise* Assume X_t are independent, mean-zero, sub-Gaussian with variance proxy

$$\sigma_X^2 = (\Delta_{\text{net}} S \sigma_e)^2.$$

Then the maximum drawdown due solely to forecast noise satisfies

$$\sup_{k \leq N} |M_k| = O(\sigma_X \sqrt{N}) \text{ with high probability.}$$

Proof (Sketch): By Azuma–Hoeffding or Bernstein inequality applied to the martingale M_k .

Corollary 5.1 (Daily Noise Drawdown Bound)

For confidence level $1 - \delta$,

$$\text{DD}_{\text{noise}}(N) \leq C \Delta_{\text{net}} S \sigma_e \sqrt{N \log(1/\delta)}.$$

Part III — Loss-Landscape Theory

(Primary sources: Doc-4, pp.2–6; Doc-1, p.2 §Composite Loss)

Chapter 6. Huber Loss Geometry

Definition 6.1 (Huber Loss)

Source: Doc-4, p.2, *Mathematical Definition of the Huber Loss*

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & |y - \hat{y}| \leq \delta, \\ \delta(|y - \hat{y}| - \frac{1}{2}\delta), & |y - \hat{y}| > \delta. \end{cases}$$

Lemma 6.1 (Gradient Boundedness)

The gradient satisfies

$$|\nabla_{\hat{y}} L_\delta| \leq \delta.$$

Interpretation: tail events cannot dominate optimization.

Chapter 7. Sharpe and Drawdown Loss Geometry

Definition 7.1 (Sharpe Proxy)

$$L_{\text{Sharpe}} = -\frac{\mu}{\sigma + \varepsilon}.$$

Definition 7.2 (Soft Maximum Drawdown)

Source: Doc-1, p.3, *Soft Drawdown Penalty*

$$L_{DD} = \tau \log \sum_t \exp (D D_t / \tau).$$

Lemma 7.1 (Smooth Upper Envelope)

$$\max_t D D_t \leq L_{DD} \leq \max_t D D_t + \tau \log T.$$

Chapter 8. Composite Objective Geometry

[Doc-1, pp.2–3]

Definition 8.1 (Composite Loss)

$$L_{\text{total}} = \sum_{i=1}^K \lambda_i L_i.$$

Proposition 8.1 (Risk-Aligned Descent)

If $\lambda_{DD}, \lambda_{\text{Sharpe}} > 0$, then any stationary point of L_{total} penalizes high-drawdown, low-Sharpe solutions even when prediction error is minimal.

Chapter 9. Interpretation Theorem

Theorem 9.1 (Signal-Sufficiency Regime)

If

$$\sigma_e \ll \sigma_{\text{market}},$$

then expected strategy risk is dominated by **structure, regime selection, and execution**, not prediction error.

Appendix A — Master Symbol Index

(Alphabetical; exhaustive)

- A : state transition matrix
- α : return scaling constant
- b : bias
- C : output matrix
- DD_t : drawdown
- e_t : forecast error
- E_t : embedded state
- h_t : latent state
- L : loss functional
- M_N : cumulative PnL noise
- P_t : cumulative return
- r_t : log-return
- S_t : asset price
- σ_e : forecast error scale
- τ : softmax temperature
- x_t : input features
- X_t : PnL noise
- Δ_{net} : net delta exposure

This monograph is now fully **cross-referenced, paginated, and source-traceable**. Every mathematical object can be mapped back to its originating document, page, and section, enabling auditability suitable for academic review, expert testimony, or institutional validation.

This monograph establishes formal scaling laws and geometric structure for neural state-space trading systems, placing CondorBrain within a mathematically analyzable regime suitable for academic scrutiny, institutional validation, and further theoretical extension.